

Systems of Equations – Word Problems

Algebra 1 (US Common Core) | Homework Practice

Method

Translate each scenario into two equations with two unknowns, then solve the system using substitution or elimination.

1 Practice Problems

1. Find the value of two numbers if their sum is 12 and their difference is 4.
2. The difference of two numbers is 3. Their sum is 13. Find the numbers.
3. Flying to Kampala with a tailwind, a plane averaged 158 km/h. On the return trip the plane averaged only 112 km/h flying back into the same wind. Find the speed of the wind and the speed of the plane in still air.
4. Stefan's school sells tickets to a choral performance. On the first day, the school sold 3 senior citizen tickets and 1 child ticket for a total of \$38. On the second day, the school took in \$52 by selling 3 senior citizen tickets and 2 child tickets. Find the price of a senior citizen ticket and the price of a child ticket.
5. The sum of the digits of a certain two-digit number is 7. Reversing its digits increases the number by 9. What is the number?
6. A boat traveled 210 miles downstream and back. The trip downstream took 10 hours; the trip back took 70 hours. Find the speed of the boat in still water and the speed of the current.
7. The senior class at High School A rented and filled 8 vans and 8 buses with 240 students. High School B rented and filled 4 vans and 1 bus with 54 students. Every van carried the same number of students, as did every bus. Find the number of students in each van and in each bus.
8. The senior class at High School A rented and filled 1 van and 6 buses with 372 students for a trip to New York City. High School B rented and filled 4 vans and 12 buses with 780 students. Each van and each bus carried the same number of students. How many students can a van carry? How many students can a bus carry?
9. Brenda's school sells tickets to a spring musical. On the first day, the school sold 3 senior citizen tickets and 9 child tickets for a total of \$75. On the second day, the school

- took in \$67 by selling 8 senior citizen tickets and 5 child tickets. Find the price of one senior citizen ticket and one child ticket.
10. Matt sold 3 small boxes of oranges and 14 large boxes of oranges for a total of \$203. Ming sold 11 small boxes and 11 large boxes for a total of \$220. Find the cost of one small box and one large box of oranges.
 11. A boat traveled 336 miles downstream and back. The trip downstream took 12 hours; the trip back took 14 hours. Find the speed of the boat in still water and the speed of the current.

2 Answer Key

1. 4 and 8
2. 5 and 8
3. Plane: 135 km/h, Wind: 23 km/h
4. Senior citizen ticket: \$8, child ticket: \$14
5. 34
6. Boat: 12 mph, Current: 9 mph
7. Van: 8 students, Bus: 22 students
8. Van: 18 students, Bus: 59 students
9. Senior citizen ticket: \$4, child ticket: \$7
10. Small box: \$7, large box: \$13
11. Boat: 26 mph, Current: 2 mph